

A clean, modal, time-sensitive logic where Recognition and Comprehension are modal operators with sharp semantics.

1. Frame and intuition

We model:

- Worlds: cognitive states over time

$$\begin{array}{l} \backslash[\\ W \ni w = \langle a, t \rangle \\ \backslash] \end{array}$$

where $\langle a \rangle$ is an agent, $\langle t \rangle$ is a time index.

- Accessibility relations:

- $(R^{\text{Rec}})_e \subseteq W \times W$: “worlds compatible with recognizing entity $\langle e \rangle$ ”
- $(R^{\text{Comp}})_e \subseteq W \times W$: “worlds compatible with comprehending entity $\langle e \rangle$ ”

Think of $(R^{\text{Rec}})_e$ as pattern-match-compatible worlds, and $(R^{\text{Comp}})_e$ as model-based-compatible worlds.

2. Syntax

2.1 Propositional base

- Atomic propositions $\langle p, q, r, \dots \rangle$
plus some distinguished atoms:

- $(\text{Model})_e$: “there exists a generative model for $\langle e \rangle$ ”
- $(\text{Predict})_e$: “predictions for $\langle e \rangle$ are accurate”
- $(\text{Track})_e$: “continuous tracking of $\langle e \rangle$ ”
- $(\text{SurpriseHigh})_e$, $(\text{SurpriseLow})_e$
- $(\text{Notice})_e$: “ $\langle e \rangle$ is consciously noticed now”

2.2 Modal operators

For each entity $\langle e \rangle$:

- $(\Box^{\text{Rec}})_e \varphi$: “in all recognition-compatible worlds for $\langle e \rangle$, φ holds”

- $\Diamond^{\text{Rec}}_e \varphi$: dual of the above
- $\Box^{\text{Comp}}_e \varphi$: “in all comprehension-compatible worlds for e , φ holds”
- $\Diamond^{\text{Comp}}_e \varphi$: dual

2.3 Formation rules (BNF)

``bnf`

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<Formula> ::= <Atom>
| "¬" <Formula>
| <Formula> "&" <Formula>
| <Formula> "∨" <Formula>
| <Formula> "→" <Formula>
| "□Rec" <Formula>
| "◇Rec" <Formula>
| "□Comp" <Formula>
| "◇Comp" <Formula>
```

(You can parametrize e explicitly or treat each e as giving a family of operators.)

3. Semantics

A model is $\langle M = \langle W, R^{\text{Rec}}_e, R^{\text{Comp}}_e, V \rangle \rangle$ where:

- W : set of worlds
- $(R^{\text{Rec}}_e, R^{\text{Comp}}_e) \subseteq W \times W$
- $V: \text{Atoms} \rightarrow \mathcal{P}(W)$: valuation

Truth definition (standard Kripke):

- $\langle M, w \rangle \models p \text{ iff } w \in V(p)$
- Boolean connectives as usual
- $\langle M, w \rangle \models \Box^{\text{Rec}}_e \varphi \text{ iff } \forall v (w R^{\text{Rec}}_e v \rightarrow \langle M, v \rangle \models \varphi)$
- $\langle M, w \rangle \models \Box^{\text{Comp}}_e \varphi \text{ iff } \forall v (w R^{\text{Comp}}_e v \rightarrow \langle M, v \rangle \models \varphi)$

4. Axiom schemas encoding your distinctions

We now encode what it means to be in recognition vs comprehension mode.

4.1 Comprehension implies model, prediction, tracking, low surprise

For each (e) :

1. Model requirement

$$\boxed{\text{Comp}}e \top \rightarrow \text{Model}e$$

2. Predictive power

$$\boxed{\text{Comp}}e \top \rightarrow \text{Predict}e$$

3. Continuous tracking

$$\boxed{\text{Comp}}e \top \rightarrow \text{Track}e$$

4. Low surprise

$$\boxed{\text{Comp}}e \top \rightarrow \text{SurpriseLow}e$$

Interpretation: if the agent is in a comprehension-compatible state for (e) , then in the actual world, model, prediction, tracking, and low surprise all hold.

4.2 Recognition without full model

1. Recognition does not guarantee model

$$\boxed{\text{Rec}}e \top \rightarrow (\neg \text{Model}e \vee \neg \text{Predict}_e)$$

2. Recognition tied to high surprise or lack of tracking

$$\boxed{\text{Rec}}e \top \rightarrow (\text{SurpriseHigh}e \vee \neg \text{Track}_e)$$

5. Mutual constraints between modalities

You can enforce that pure comprehension and pure recognition are incompatible modes for the same entity at the same world:

1. No simultaneous pure recognition and comprehension

$$\begin{array}{l} \backslash[\\ \Box^{\{\text{Comp}\}}e \top \rightarrow \neg \Box^{\{\text{Rec}\}}e \top \\ \backslash] \end{array}$$

Optionally also:

$$\begin{array}{l} \backslash[\\ \Box^{\{\text{Rec}\}}e \top \rightarrow \neg \Box^{\{\text{Comp}\}}e \top \\ \backslash] \end{array}$$

(You can relax this if you want “mixed” states.)

6. Noticing as a modal event

We capture your key invariant: noticing implies prior non-comprehension.

Introduce a temporal predecessor relation on worlds, $\backslash(\text{Prev})(w', w)\backslash$, and treat it meta-logically, or encode it as a binary predicate if you extend the language.

A schema:

- Noticing implies high surprise:

$$\begin{array}{l} \backslash[\\ \text{Notice}e \rightarrow \text{SurpriseHigh}e \\ \backslash] \end{array}$$

- Noticing implies absence of stable comprehension just before:

Informally: if at world $\backslash(w)\backslash$ we have $\backslash(\text{Notice})_e\backslash$, then at the immediately preceding world $\backslash(w')\backslash$, comprehension modality for $\backslash(e)\backslash$ was not active:

$$\begin{array}{l} \backslash[\\ \text{forall } w, w'. \backslash(\text{Prev})(w', w) \wedge M, w \vdash \text{Notice}_e \\ \rightarrow M, w' \nvdash \Box^{\{\text{Comp}\}}_e \top \\ \backslash] \end{array}$$

If you want this inside the object language, you can move to a temporal–modal fusion (e.g., add a past operator $\backslash(P\backslash)$):

- Add operator $\backslash(P\backslash)$ (“previous moment”).
- Then:

$$\backslash[\backslash\mathrm{Notice}e \rightarrow P(\neg \Box^{\backslash\mathrm{Comp}}e \top)\backslash]$$

This is exactly your insight: if I notice it now, then just before now I was not in a comprehension state for it.

7. Transition principles (as modal-temporal laws)

You can express upgrades and degradations:

- Upgrade (Recognition $\rightarrow$ Comprehension):

$$\backslash[(\Box^{\backslash\mathrm{Rec}}e \top \wedge \backslash\mathrm{Model}e \wedge \backslash\mathrm{Predict}e \wedge \backslash\mathrm{Track}e \wedge \backslash\mathrm{SurpriseLow}_e) \rightarrow F(\Box^{\backslash\mathrm{Comp}}e \top)\backslash]$$

where $\backslash(F\backslash)$ is a future operator (“eventually”).

- Degradation (Comprehension $\rightarrow$ Recognition):

$$\backslash[(\Box^{\backslash\mathrm{Comp}}e \top \wedge \backslash\mathrm{SurpriseHigh}e) \rightarrow F(\Box^{\backslash\mathrm{Rec}}e \top)\backslash]$$